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In [16]: import nltk
from nltk import word_tokenize, pos_tag
from collections import Counter

nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
tokens = word_tokenize(medical_text)
print("Tokens:")
print(tokens)
pos_tags = pos_tag(tokens)
print("\nPOS Tags:")
print(pos_tags)
nouns = [word for word, tag in pos_tags if tag.startswith('NN')]
verbs = [word for word, tag in pos_tags if tag.startswith('VB')]

print("\nNouns:", nouns)
print("Verbs:", verbs)
print("\nNoun Frequency:", Counter(nouns))
print("Verb Frequency:", Counter(verbs))
```

Tokens:

```
['Diabetes', 'is', 'a', 'chronic', 'disease', 'that', 'affects', 'how', 'the',
'body', 'processes', 'blood', 'sugar', '.', 'If', 'untreated', ',', 'diabetes',
'may', 'cause', 'heart', 'disease', ',', 'kidney', 'failure', ',', 'nerve', 'da
mage', 'and', 'vision', 'problems', '.', 'Early', 'diagnosis', 'and', 'proper',
'treatment', 'help', 'improve', 'patient', 'outcomes', '.']
```

POS Tags:

```
[('Diabetes', 'NNS'), ('is', 'VBZ'), ('a', 'DT'), ('chronic', 'JJ'), ('diseas
e', 'NN'), ('that', 'WDT'), ('affects', 'VBZ'), ('how', 'WRB'), ('the', 'DT'),
('body', 'NN'), ('processes', 'VBZ'), ('blood', 'NN'), ('sugar', 'NN'), ('.',
'), ('If', 'IN'), ('untreated', 'JJ'), (',', ','), ('diabetes', 'NNS'), ('ma
y', 'MD'), ('cause', 'VB'), ('heart', 'NN'), ('disease', 'NN'), (',', ','), ('k
idney', 'NN'), ('failure', 'NN'), (',', ','), ('nerve', 'NN'), ('damage', 'N
N'), ('and', 'CC'), ('vision', 'NN'), ('problems', 'NNS'), ('.', '.'), ('Earl
y', 'JJ'), ('diagnosis', 'NN'), ('and', 'CC'), ('proper', 'JJ'), ('treatment',
'NN'), ('help', 'NN'), ('improve', 'VB'), ('patient', 'NN'), ('outcomes', 'NN
S'), (',', ',')]
```

Nouns: ['Diabetes', 'disease', 'body', 'blood', 'sugar', 'diabetes', 'heart', 'disease', 'kidney', 'failure', 'nerve', 'damage', 'vision', 'problems', 'diagnosis', 'treatment', 'help', 'patient', 'outcomes']

Verbs: ['is', 'affects', 'processes', 'cause', 'improve']

Noun Frequency: Counter({'disease': 2, 'Diabetes': 1, 'body': 1, 'blood': 1, 'sugar': 1, 'diabetes': 1, 'heart': 1, 'kidney': 1, 'failure': 1, 'nerve': 1, 'damage': 1, 'vision': 1, 'problems': 1, 'diagnosis': 1, 'treatment': 1, 'help': 1, 'patient': 1, 'outcomes': 1})

Verb Frequency: Counter({'is': 1, 'affects': 1, 'processes': 1, 'cause': 1, 'improve': 1})

```
[nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data]   Package punkt is already up-to-date!  
[nltk_data] Downloading package averaged_perceptron_tagger to  
[nltk_data]   /root/nltk_data...  
[nltk_data]   Package averaged_perceptron_tagger is already up-to-  
[nltk_data]   date!
```